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main.py

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```
1 points=[(1,2),(4,5),(7,8),(3,1)]
2 length=len(points)
3 min_distance=float('inf')
4 closest_pair=0
5 for i in range(length):
6     for j in range(i+1,length):
7         p1=points[i]
8         p2=points[j]
9         distance=((p1[0]-p2[0])**2+(p1[1]-p2[1])**2)**0.5
10        if(distance<min_distance):
11            min_distance=distance
12            closest_pair=(p1,p2)
13 print("closest pair:",p1,",",p2)
14 print("minimum distance:",min_distance)
```

Output

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closest pair: (7, 8) , (3, 1)
minimum distance: 2.23606797749979

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10        if(distance<min_distance):
11            min_distance=distance
12            closest_pair=(p1,p2)
13 print("closest pair:",p1,"",p2)
14 print("minimum distance:",min_distance)
15
16 points=[(10,0),(11,5),(5,3),(9,3.5),(15,3),(12.5,7),(6,6.5),(7.5,4.5
    )]
17 n=len(points)
18 hull=[]
19 leftmost=0
20 for i in range(1,n):
21     if(points[i][0]<points[leftmost][0]):
```

Output

Clear

closest pair: (7, 8) , (3, 1)
minimum distance: 2.23606797749979
convex hull:
(5, 3)
(10, 0)
(15, 3)
(12.5, 7)
(6, 6.5)

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```
17 n=len(points)
18 hull=[]
19 leftmost=0
20 for i in range(1,n):
21     if(points[i][0]<points[leftmost][0]):
22         leftmost=i
23 p=leftmost
24 while(True):
25     hull.append(points[p])
26     q=(p+1)%n
27     for i in range(n):
28         val=(points[i][1]-points[p][1])*(points[q][0]-points[i][0])
29         -(points[i][0]-points[p][0])*(points[q][1]-points[i][1])
30         if(val<0):
31             q=i
32     if(p==leftmost):
33         break
34 print("convex hull:")
35 for point in hull:
36     print(point)
```

Output

Clear

closest pair: (7, 8) , (3, 1)
minimum distance: 2.23606797749979
convex hull:
(5, 3)
(10, 0)
(15, 3)
(12.5, 7)
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```
1 cities = [(1, 2), (4, 5), (7, 1), (3, 6)]
2 start = cities[0]
3 remaining_cities = [cities[1], cities[2], cities[3]]
4 routes = [
5     [remaining_cities[0], remaining_cities[1], remaining_cities[2]],
6     [remaining_cities[0], remaining_cities[2], remaining_cities[1]],
7     [remaining_cities[1], remaining_cities[0], remaining_cities[2]],
8     [remaining_cities[1], remaining_cities[2], remaining_cities[0]],
9     [remaining_cities[2], remaining_cities[0], remaining_cities[1]],
10    [remaining_cities[2], remaining_cities[1], remaining_cities[0]]
11 ]
12 min_distance = 1000000
13 best_route = []
14 for route in routes:
15     route = [start] + route + [start]
16     distance = 0
17     for i in range(len(route) - 1):
18         X1, Y1 = route[i]
19         X2, Y2 = route[i + 1]
20         distance += ((X2 - X1) ** 2 + (Y2 - Y1) ** 2) ** 0.5
21     if distance < min_distance:
22         min_distance = distance
```

Output

Clear

Shortest distance: 16.969112047670894
Shortest route: [(1, 2), (7, 1), (4, 5), (3, 6), (1, 2)]

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```
1 cost_matrix = [[3, 10, 7],[8,5,12],[4,6,9]]
2 n = 3
3 optimal_cost = float('inf')
4 optimal_assignment = []
5 permutations = [
6     [0, 1, 2], [0, 2, 1], [1, 0, 2],
7     [1, 2, 0], [2, 0, 1], [2, 1, 0]
8 ]
9 for perm in permutations:
10     total_cost = (
11         cost_matrix[0][perm[0]] +
12         cost_matrix[1][perm[1]] +
13         cost_matrix[2][perm[2]]
14     )
15     if total_cost < optimal_cost:
16         optimal_cost = total_cost
17         optimal_assignment = perm
18 print("Optimal assignment:")
19 print("worker 1, task:", optimal_assignment[0] + 1)
20 print("worker 2, task:", optimal_assignment[1] + 1)
21 print("worker 3, task:", optimal_assignment[2] + 1)
22 print("Total cost:", optimal_cost)
```

Output

Clear

Optimal assignrnt:
worker 1, task: 3
worker 2, task: 2
worker 3, task: 1
Total cost: 16

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Output

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```
1 weight=[2,3,1]
2 values=[4,5,3]
3 capacity=4
4 n=len(weight)
5 max_value=0
6 best_combination=[]
7 for i in range(1<=n):
8     combination=[]
9     total_weight=0
10    total_value=0
11    for j in range(n):
12        if(i and (1<=j)):
13            combination.append(j)
14            total_weight+=weight[j]
15            total_value+=values[j]
16    if(total_weight<=capacity and total_value>max_value):
17        max_value=total_value
18        best_combination=combination
19    print("optimal selection:",best_combination)
20    print("total value:",max_value)
```

optimal selection: []

total value: 0

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